

A Review of Image Analysis and Inpainting Techniques, and Motivation for DiffIndEye

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Abstract

This project studies retinal Optical Coherence Tomography (OCT) inpainting using both classical image reconstruction methods and diffusion-based generative modeling. The framework reframes retinal OCT reconstruction as a conditional inverse problem in which missing anatomical structure must be reconstructed from partially observed retinal scans. A complete experimental pipeline was developed including image preprocessing, synthetic stripe corruption, classical inpainting baselines, conditional denoising diffusion probabilistic models (DDPMs), and modular evaluation workflows. Classical reconstruction methods including Gaussian smoothing, PDE-based diffusion, and Telea inpainting were compared against both an initial encoder-decoder DDPM and a lightweight U-Net diffusion architecture. Experimental results showed that classical methods, particularly Telea inpainting, substantially outperformed the initial diffusion implementations despite their greater model complexity. Although the diffusion models trained stably and successfully optimized the denoising objective, they struggled to preserve detailed retinal anatomy and long-range structural continuity. These findings suggest that successful retinal OCT reconstruction depends critically on architectural design, multi-scale spatial feature propagation, and anatomically meaningful inductive bias in addition to the diffusion framework itself. The project ultimately establishes DiffIndEye as a reproducible experimental framework for future research in diffusion-based medical image reconstruction, uncertainty-aware retinal imaging, and probabilistic anatomical inpainting.

Keywords: retinal OCT, medical image analysis, image inpainting, diffusion models, DDPM, computer vision, generative modeling, retinal reconstruction, medical imaging, deep learning, probabilistic reconstruction, computational ophthalmology

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1 Introduction

Medical image analysis studies how quantitative methods can be used to represent, process, and interpret images arising from biological and clinical systems. In retinal optical coherence tomography (OCT), the image is not merely a visual object but a structured measurement of tissue reflectivity. This makes OCT reconstruction a mathematical imaging problem: one must preserve geometry, intensity, and anatomical plausibility while dealing with noise, artifacts, and missing information.

The goal of this report is to review the image analysis ideas underlying retinal OCT inpainting and to motivate DiffIndEye, an exploratory framework for diffusion-based inpainting of retinal OCT images. The project begins with classical reconstruction methods, including zero-fill, Gaussian smoothing, PDE-style diffusion, and Telea inpainting, before moving to conditional denoising diffusion models. The central question is whether a learned generative model can recover plausible retinal structure from synthetically corrupted OCT scans, and how this compares to simpler deterministic baselines.

Medical Image Analysis and Computer Vision

A grayscale medical image can be represented as a function $I : \Omega \rightarrow \mathbb{R}$, where $\Omega \subset \mathbb{R}^2$ is the spatial image domain and $I(x, y)$ is the measured intensity at location (x, y) . After discretization, the image is stored as a matrix $I \in \mathbb{R}^{m \times n}$. Computer vision methods then attempt to extract structure from this matrix, such as edges, regions, textures, shapes, or higher-level semantic information.

Medical image analysis differs from ordinary computer vision because the structures being analyzed are constrained by anatomy and imaging physics. Small changes in intensity or geometry may correspond to clinically meaningful tissue differences. Therefore, image processing methods must balance numerical accuracy with structural plausibility.

Many medical imaging problems can be written as inverse problems. If X denotes the unknown clean image and Y denotes the observed image, then a general degradation model is

$$Y = \mathcal{A}(X) + \varepsilon,$$

where $\mathcal{A}$ is an acquisition or corruption operator and ε represents noise. Denoising, deblurring, super-resolution, segmentation, and inpainting can all be viewed as attempts to recover or infer information about X from Y .

Retinal Imaging and Optical Coherence Tomography

Optical coherence tomography is a non-invasive imaging modality that produces cross-sectional views of retinal tissue. An OCT B-scan visualizes layered retinal anatomy by measuring reflected light intensity at different depths. In the resulting image, bright regions often

correspond to highly reflective tissue boundaries, while dark regions may correspond to fluid, attenuation, or low-reflectivity tissue.

Retinal OCT images are highly structured. Unlike natural images, they usually contain strong horizontal layer organization, smooth anatomical continuity, and disease-specific distortions. Conditions such as age-related macular degeneration, diabetic macular edema, choroidal neovascularization, and drusen alter retinal shape and reflectivity. This makes OCT imaging useful for diagnosis, but also challenging for reconstruction because missing or corrupted regions may contain clinically important structure.

Image Corruption and Missing Information Problems

Image corruption occurs when the observed image fails to fully capture the underlying clean structure. In OCT imaging, this can arise from speckle noise, motion artifacts, signal attenuation, shadowing, poor fixation, or acquisition dropout. For inpainting, the relevant case is partial observation.

Let $X \in \mathbb{R}^{H \times W}$ denote a clean image and let $M \in \{0, 1\}^{H \times W}$ denote a binary mask. The observed corrupted image is

$$Y = M \odot X,$$

where $\odot$ denotes elementwise multiplication. Pixels with $M_{ij} = 1$ are observed, while pixels with $M_{ij} = 0$ are missing. The inpainting task is to estimate the missing values using the observed image Y and mask M .

This problem is ill-posed because many different images may agree with the observed pixels. A deterministic method returns one reconstruction $\hat{X}$, while a probabilistic method attempts to model a conditional distribution $p(X | Y, M)$ over plausible reconstructions.

Motivation for Retinal OCT Inpainting

Retinal OCT inpainting is motivated by the need to recover or approximate missing anatomical structure in corrupted scans. Because OCT images contain layered retinal tissue, inpainting is not just a pixel-filling problem. A successful reconstruction should preserve layer continuity, local reflectivity patterns, and global retinal shape.

Classical inpainting methods can be effective when missing regions are small or when local continuity is sufficient. However, structured missing regions may require more than local interpolation. In retinal OCT, a missing horizontal band can remove part of several retinal layers simultaneously, forcing the reconstruction method to infer anatomy from broader context.

1.1 Motivation for DiffIndEye

DiffIndEye was developed to study retinal OCT inpainting using both classical image processing baselines and diffusion-based generative modeling. The project asks whether conditional diffusion models can learn retinal structure well enough to reconstruct synthetically masked OCT regions.

The motivation is twofold. First, diffusion models provide a probabilistic framework for image reconstruction, where multiple plausible completions can in principle be sampled from $p(X | Y, M)$. Second, comparing diffusion models against classical methods reveals whether the added complexity of generative modeling is justified for this task.

The project also emphasizes reproducibility. The pipeline separates preprocessing, masking, baselines, diffusion schedules, model training, sampling, and evaluation into modular components so that future diffusion architectures can be tested systematically.

1.2 Project Objectives

The objectives of DiffIndEye are:

preprocess OCT images $\rightarrow$ apply controlled masks $\rightarrow$ compare reconstruction methods.

More specifically, the project aims to construct a reproducible OCT inpainting framework, benchmark classical inpainting methods, implement a conditional denoising diffusion model, evaluate reconstruction quality under synthetic stripe corruption, and identify limitations of the initial diffusion architecture.

Research Questions

The project is guided by the following questions. Can a diffusion model learn meaningful retinal OCT structure from masked images? How does diffusion-based reconstruction compare with classical baselines such as Gaussian smoothing, PDE diffusion, and Telea inpainting? Why might a diffusion model train successfully but still reconstruct poorly? What architectural changes, such as a U-Net backbone, are likely necessary for better OCT inpainting? Finally, can this framework be extended toward uncertainty-aware reconstruction and more realistic retinal artifact correction?

2 Background Review

Foundations of Image Analysis

Image analysis studies how mathematical methods can be used to represent, process, and interpret visual information. As stated in the introduction, a grayscale image is commonly modeled as a function

$$I : \Omega \rightarrow \mathbb{R},$$

where $\Omega \subset \mathbb{R}^2$ denotes the image domain and $I(x, y)$ is the image intensity at spatial coordinate (x, y) . After discretization, the image becomes a matrix

$$I \in \mathbb{R}^{m \times n}.$$

Many image analysis tasks rely on extracting structure from this matrix representation. Edges correspond to regions of rapid intensity variation and are often characterized using image gradients:

$$\nabla I = \left(\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y} \right).$$

The gradient magnitude

$$\|\nabla I\| = \sqrt{\left(\frac{\partial I}{\partial x} \right)^2 + \left(\frac{\partial I}{\partial y} \right)^2}$$

is frequently used to identify boundaries and transitions in anatomical structure. Image analysis also studies spatial correlation and texture. Neighboring pixels in medical images are usually highly dependent, meaning:

$$\text{Cov}(I_{ij}, I_{kl}) > 0$$

for nearby spatial locations. This local structure motivates convolutional methods, diffusion processes, and neighborhood-based reconstruction techniques.

Classical Image Processing Approaches

Classical image processing methods reconstruct images using local intensity relationships rather than learned statistical models. These approaches are often computationally efficient and mathematically interpretable.

One of the simplest reconstruction ideas is interpolation. If neighboring pixel values are known, missing pixels can be estimated as weighted averages:

$$\hat{I}(x, y) = \sum_k w_k I_k.$$

Gaussian smoothing is another classical operation. A Gaussian kernel

$$G_{\sigma}(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right)$$

is convolved with the image to reduce noise and smooth discontinuities. These methods work well when missing regions are small and local continuity assumptions hold. However, they often oversmooth boundaries and fail to reconstruct large missing structures. In retinal OCT imaging, excessive smoothing can destroy thin retinal layers and clinically important reflectivity transitions.

Partial Differential Equation Methods

Partial differential equation methods interpret image reconstruction as a continuous diffusion process. The simplest PDE model is isotropic diffusion:

$$\frac{\partial u}{\partial t} = \Delta u,$$

where Δ is the Laplacian operator. This equation smooths intensity differences over time and can be interpreted as heat flow across the image domain. Anisotropic diffusion modifies the diffusion strength using local image gradients:

$$\frac{\partial u}{\partial t} = \nabla \cdot (c(\|\nabla u\|)\nabla u),$$

where $c(\cdot)$ controls diffusion strength. Regions with large gradients experience less smoothing, which helps preserve edges. PDE methods are attractive because they are mathematically grounded and preserve local continuity better than naive smoothing. However, they remain fundamentally local methods. Large missing regions often require global structural inference, which PDE diffusion alone cannot provide.

Patch-Based and Exemplar Methods

Patch-based inpainting reconstructs missing regions by copying similar patches from observed regions of the image. Instead of interpolating locally, these methods search for image regions with compatible texture and structure. Let P_t denote a target patch centered in a missing region and let P_s denote a candidate source patch elsewhere in the image. Reconstruction seeks:

$$P_s^* = \arg \min_{P_s} d(P_t, P_s),$$

where $d(\cdot, \cdot)$ is a similarity measure.

Patch-based methods can preserve texture and repetitive structure effectively. However, they assume that suitable replacement patches already exist within the image. In retinal OCT scans, pathology and anatomical uniqueness may violate this assumption. Patch copying can also introduce discontinuities or unrealistic anatomical repetition.

Deep Learning for Image Reconstruction

Deep learning approaches learn image reconstruction directly from data. Convolutional neural networks process images using learned spatial filters:

$$h_{ij}^{(k)} = \sigma \left(\sum_{u,v} W_{uv}^{(k)} x_{i+u,j+v} + b^{(k)} \right),$$

where W denotes convolution kernels and σ is a nonlinear activation function. Encoder-decoder architectures compress images into latent representations before reconstructing them. U-Net architectures extend this idea using skip connections that preserve high-resolution spatial detail. Deep learning methods can model highly nonlinear image structure and often outperform classical interpolation methods. However, deterministic reconstruction models still produce only one reconstruction estimate and may hallucinate unrealistic anatomy. Performance also depends strongly on architecture design and training stability.

Diffusion Models in Computer Vision

Diffusion probabilistic models generate images through iterative denoising. A forward process gradually corrupts an image with Gaussian noise:

$$q(x_t | x_{t-1}) = \mathcal{N} \left(\sqrt{1 - \beta_t} x_{t-1}, \beta_t I \right),$$

where β_t controls the variance schedule. After many diffusion steps, the image approaches Gaussian noise. The model then learns a reverse process that reconstructs the image:

$$p_\theta(x_{t-1} | x_t).$$

A common formulation predicts the added noise directly:

$$\mathcal{L} = \mathbb{E} \left[\|\varepsilon - \varepsilon_\theta(x_t, t)\|^2 \right].$$

Diffusion models have become important in computer vision because they provide stable generative training and strong sample quality. Unlike GANs, they model probability distributions explicitly and support iterative reconstruction.

Medical Imaging Applications of Diffusion Models

Diffusion models have recently been applied to denoising, reconstruction, segmentation, synthesis, and super-resolution tasks in medical imaging. In MRI and CT imaging, diffusion models have been used to reconstruct missing or undersampled measurements. In pathology and microscopy, they have been applied to generative image synthesis and enhancement.

Medical imaging presents unique challenges for generative models because anatomy imposes strong structural constraints. Reconstructions must preserve geometry, tissue boundaries, and clinically meaningful morphology. This makes probabilistic reconstruction especially appealing, since multiple plausible reconstructions may exist for partially observed medical images.

In retinal OCT imaging, diffusion models are attractive because retinal anatomy exhibits strong spatial organization and repeated layer structure. However, preserving fine retinal boundaries and pathological variation remains difficult.

Limitations of Existing Inpainting Approaches

Existing inpainting approaches exhibit several important limitations. Classical interpolation and smoothing methods oversimplify image structure and struggle with large missing regions. PDE methods preserve local continuity but remain fundamentally local. Patch-based methods depend heavily on self-similarity within the image and may fail when pathology is unique.

Deep learning methods improve reconstruction quality but may hallucinate anatomically implausible structure. GAN-based models can generate sharp images but are difficult to train and may collapse to limited image modes.

Diffusion models address some of these issues through probabilistic iterative denoising. However, they are computationally expensive and highly sensitive to architecture design, noise schedules, and conditioning strategies. A model may optimize its denoising objective successfully while still producing poor anatomical reconstruction.

These limitations motivate the study of diffusion-based retinal OCT reconstruction within a controlled experimental framework. DiffIndEye was developed specifically to compare classical deterministic baselines against diffusion-based probabilistic reconstruction while identifying the architectural bottlenecks that currently limit OCT inpainting performance.

3 Dataset and Exploratory Analysis

3.1 Retinal OCT Dataset Description

The DiffIndEye framework uses a publicly available retinal optical coherence tomography dataset consisting of grayscale retinal B-scans acquired from multiple disease categories. OCT imaging produces cross-sectional measurements of retinal tissue reflectivity, allowing layered retinal anatomy to be visualized in high resolution.

The dataset contains approximately 18,400 training images and 2,800 test images. Images were originally collected for retinal disease classification tasks but are repurposed here for reconstruction and inpainting experiments. Each image can be represented as $I \in \mathbb{R}^{m \times n}$, where

pixel intensity corresponds to retinal reflectivity. Brighter intensities usually indicate highly reflective tissue interfaces, while darker regions correspond to low reflectivity or attenuated signal. Unlike natural image datasets, retinal OCT scans exhibit highly organized anatomical structure. Most images contain horizontally aligned retinal layers with smooth spatial continuity and disease-dependent structural variation. This strong geometric organization makes OCT imaging particularly suitable for structured reconstruction experiments.

Class Structure and Dataset Composition

The dataset contains four major retinal categories:

$$\{\text{CNV}, \text{DME}, \text{Drusen}, \text{Normal}\}.$$

Choroidal neovascularization (CNV) images often exhibit abnormal vascular growth, fluid accumulation, and severe structural distortion. Diabetic macular edema (DME) scans frequently contain cystic fluid regions and retinal swelling. Drusen images contain deposits associated with dry age-related macular degeneration and often exhibit gradual deformation of retinal boundaries. Normal images preserve relatively regular retinal layer organization.

Although the present project does not perform disease classification, the class diversity is important because it increases anatomical variability within the learned image distribution:

$$p_{\text{data}}(x).$$

The reconstruction framework therefore learns from both healthy and pathological retinal morphology. This produces a more difficult and realistic reconstruction problem than learning from healthy anatomy alone.

Image Characteristics

Retinal OCT images possess several structural properties that distinguish them from ordinary natural images. Most importantly, retinal scans exhibit strong horizontal anatomical organization. Thin retinal layers extend laterally across the image, creating high spatial correlation and continuous geometric structure.

After preprocessing, each image is resized to 128×256 , giving tensor representations of the form

$$X \in \mathbb{R}^{1 \times 128 \times 256}.$$

Images are grayscale rather than RGB, meaning that anatomical information is encoded primarily through intensity and spatial structure rather than color. This makes OCT reconstruction fundamentally a structural intensity reconstruction problem.

Pathological scans introduce substantial variability into the dataset. Fluid accumulation, reflectivity changes, retinal thickening, and irregular anatomical boundaries all alter local image statistics. Consequently, successful reconstruction methods must preserve both local retinal texture and global anatomical organization.

3.2 Exploratory Visualization

Exploratory visualization was performed to understand retinal anatomy, image variability, and reconstruction difficulty before model development. Random image sampling revealed that retinal OCT images consistently contain horizontally layered anatomy, strong spatial continuity, localized pathological deformation, and high contrast tissue boundaries. Visualization across disease categories demonstrated substantial structural heterogeneity. CNV and DME scans frequently contain fluid disruptions and irregular morphology, while normal scans exhibit smoother layer continuity.

Exploratory inspection also motivated the use of horizontal stripe masking. Because retinal layers extend laterally across the image, removing a horizontal band interrupts multiple retinal layers simultaneously, creating a structured and anatomically meaningful reconstruction problem. Visual inspection of masked images confirmed that stripe corruption removes clinically relevant retinal structure while preserving enough surrounding anatomy to make reconstruction feasible.

Intensity and Structural Analysis

Retinal OCT images exhibit highly structured intensity distributions. Bright regions correspond to reflective retinal interfaces, while darker regions often correspond to fluid or attenuated signal.

The image gradient

$$\nabla I = \left(\frac{\partial I}{\partial x}, \frac{\partial I}{\partial y} \right)$$

is especially informative because retinal layer boundaries produce strong local intensity transitions. Large gradient magnitudes therefore frequently correspond to clinically meaningful anatomical interfaces. The dataset also exhibits strong spatial correlation:

$$\text{Cov}(I_{ij}, I_{kl}) > 0$$

for nearby spatial coordinates. This local continuity motivates convolutional architectures and PDE-based smoothing methods. At the same time, retinal OCT images contain both low-frequency and high-frequency structure. Global retinal curvature and large-scale anatomy correspond to low-frequency structure, while thin retinal boundaries and pathological detail correspond to high-frequency information. Successful reconstruction therefore requires preserving multiple spatial scales simultaneously.

Observed Imaging Artifacts and Variability

Several forms of imaging variability and artifact were observed during exploratory analysis. Speckle noise appears throughout many scans due to coherent interference effects during OCT acquisition. This produces granular local intensity variation that can obscure retinal boundaries. Motion artifacts occasionally introduce discontinuities or local distortion. Signal attenuation and vascular shadowing may also produce darker vertical regions that partially obscure deeper retinal tissue. Pathological variation introduces an additional source of complexity. Fluid regions, retinal swelling, drusen deposits, and irregular layer deformation all alter anatomical structure in ways that may complicate reconstruction. Contrast and brightness also vary substantially across scans due to acquisition conditions and scanner preprocessing. Consequently, intensity normalization and standardized preprocessing become important for stable diffusion training.

These observations motivated the design of a controlled reconstruction framework in which corruption patterns are generated synthetically while preprocessing remains standardized across all experiments.

4 Image Preprocessing Pipeline

Preprocessing Objectives

Image preprocessing was designed to standardize retinal OCT scans before reconstruction experiments and diffusion model training. The dataset contains substantial variability in resolution and contrast. Without preprocessing, diffusion models may incorrectly learn scanner-dependent artifacts or unstable intensity distributions rather than retinal anatomical structure. The pipeline was also designed to support reproducible experimentation. All preprocessing operations were deterministic and integrated into reusable dataset and training utilities.

Image Loading and Grayscale Conversion

Retinal OCT images were loaded directly from the dataset directory structure using a modular PyTorch-based dataset pipeline. Each image was converted into grayscale format before further processing. Grayscale representation is appropriate for OCT imaging because anatomical information is encoded primarily through reflectivity and boundary structure, rather than color. After loading, each image is converted into a tensor representation where the leading dimension corresponds to the grayscale channel.

Image Resizing and Standardization

Original OCT scans vary in spatial resolution and aspect ratio. To ensure consistent tensor dimensions and stable convolutional processing, all images were resized to:

$$\mathcal{T} : \mathbb{R}^{m \times n} \rightarrow \mathbb{R}^{128 \times 256}.$$

This standardization simplifies batch processing, mask generation, and other downstream analyses. The selected resolution balances anatomical fidelity against computational feasibility. Larger images preserve more retinal detail but increase training cost substantially, while smaller images risk destroying thin retinal boundaries and pathological structure. Because retinal OCT images possess strong horizontal organization, preserving approximate aspect ratio was important for maintaining retinal geometry during resizing.

Intensity Normalization

Retinal OCT scans exhibit substantial variation in brightness and contrast due to acquisition settings and pathological differences. To stabilize optimization, image intensities were normalized into the interval (0,1). For image I , normalization is performed using Min-Max Normalization:

$$I_{\text{norm}} = \frac{I - I_{\min}}{I_{\max} - I_{\min}}.$$

Normalization improves numerical stability during diffusion training because the forward diffusion process injects Gaussian noise directly into image tensors:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\epsilon.$$

Without normalization, inconsistent intensity scales would alter the effective signal-to-noise ratio across images and destabilize denoising behavior. Normalization also ensures consistent evaluation metrics across experiments. Quantities such as mean squared error and PSNR become comparable only when images share a common intensity range.

Tensor Preparation for Deep Learning

After preprocessing, retinal OCT images were converted into tensors compatible with convolutional neural networks and diffusion models. During training, images are processed in batches:

$$X \in \mathbb{R}^{B \times 1 \times 128 \times 256},$$

where B denotes batch size. Synthetic corruption masks are represented as binary tensors:

$$M \in \{0, 1\}^{1 \times 128 \times 256}.$$

The masked retinal image is then computed as

$$Y = M \odot X.$$

The conditional diffusion model receives (X_t, Y, M) where X_t is the noisy diffusion state, Y is the masked retinal image, and M is the corruption mask. These tensors are concatenated along the channel dimension before convolutional processing. In the baseline implementation, the model therefore operates on three-channel conditional inputs consisting of the noisy image, the mask, and the masked image.

Pipeline Validation and Sanity Checks

Several validation procedures were performed to ensure preprocessing consistency and training stability.

Dataset loading checks verified that training, validation, and testing were correctly discovered and processed.

Tensor shape validation confirmed:

$$\begin{aligned} \text{image} &\sim (B, 1, 128, 256), \\ \text{mask} &\sim (B, 1, 128, 256), \\ \text{masked_image} &\sim (B, 1, 128, 256). \end{aligned}$$

Normalization validation ensured that image intensities satisfied $X_{ij} \in [0, 1]$.

Visual inspection was also used to confirm that retinal anatomy remained visible after preprocessing, stripe masks removed meaningful retinal structure, and tensor construction behaved consistently across samples. Diffusion schedule sanity checks verified that $\bar{\alpha}_0 \approx 1$ and $\bar{\alpha}_T \approx 0$, ensuring that early diffusion states preserve image signal while late states approach near-complete Gaussian noise. These validation procedures established a stable and reproducible preprocessing framework for subsequent reconstruction experiments.

5 Synthetic Corruption and Masking

Inpainting Problem Formulation

Let $X \in \mathbb{R}^{H \times W}$ denote a clean retinal OCT image and let $M \in \{0, 1\}^{H \times W}$ denote a binary mask indicating observed and missing pixels. The corrupted image is defined as $Y = M \odot X$, where $\odot$ denotes elementwise multiplication. Pixels satisfying $M_{ij} = 1$ remain visible, while pixels satisfying $M_{ij} = 0$ are removed. The inpainting problem consists

of estimating a reconstruction $\hat{X} \approx X$ using only the corrupted image Y and the mask M . From a probabilistic perspective, reconstruction can be interpreted as estimating the conditional distribution $p(X | Y, M)$. This formulation is naturally ill-posed because multiple plausible retinal structures may agree with the observed pixels. In retinal OCT imaging, the reconstruction problem is particularly challenging because corruption may interrupt multiple retinal layers simultaneously.

Motivation for Synthetic Masking

Naturally occurring OCT artifacts are difficult to standardize because they vary substantially across scanners, acquisition conditions, and disease states. As a result, evaluating reconstruction methods directly on real artifacts makes controlled benchmarking difficult. Synthetic masking addresses this issue by providing reproducible corruption geometry, known ground-truth anatomy, and controlled reconstruction difficulty. Since the original retinal image remains available, reconstruction quality can be evaluated quantitatively using metrics such as MSE, MAE, and PSNR.

Synthetic corruption also allows direct comparison between deterministic interpolation, Heat-inspired information diffusion, and generative reconstruction. The objective is not necessarily to mimic one specific OCT artifact exactly, but rather to create structured missing regions that challenge the model’s ability to reconstruct anatomically plausible retinal structure.

Horizontal Stripe Corruption Design

The corruption strategy used in DiffIndEye consists of horizontal stripe masking. This design was motivated by the layered geometry of retinal OCT images, where anatomical structure extends predominantly across the horizontal direction.

A stripe mask removes a contiguous horizontal band: $[y_0, y_0 + h)$, where y_0 is the starting row coordinate, and h is the stripe height. The mask is therefore defined by

$$M_{ij} = \begin{cases} 0 & y_0 \leq i < y_0 + h, \\ 1 & \text{otherwise.} \end{cases}$$

This corruption pattern removes portions of multiple retinal layers simultaneously. Unlike random sparse masking, stripe corruption disrupts long-range anatomical continuity and creates a substantially more difficult reconstruction problem.

5.1 Mask Generation Procedure

Mask generation was implemented procedurally during preprocessing. For each retinal OCT image:

1. a stripe height h is selected,
2. a vertical starting coordinate y_0 is sampled,
3. a binary mask M is constructed,
4. and the corrupted image $Y = M \odot X$ is computed.

Because masks are generated dynamically, the framework can easily support future corruption strategies such as multiple stripe regions and variable stripe widths.

Mask Parameterization

The primary mask parameter is stripe height h . Increasing h increases reconstruction difficulty because larger portions of retinal anatomy are removed. Small stripe heights preserve substantial local context, making interpolation relatively straightforward, while larger masks require more global structural inference.

The stripe position parameter y_0 also influences reconstruction complexity. Different retinal regions exhibit different structural characteristics and reflectivity patterns. Corrupting highly reflective retinal boundaries or pathological regions may produce substantially more difficult reconstruction tasks.

The stripe masks span the full image width $0 \leq j < W$. This forces reconstruction methods to recover global anatomical structure rather than simply interpolating isolated local pixels. The current implementation uses fixed stripe geometry for experimental consistency, although the framework remains extensible to more complex corruption schemes.

5.2 Visual Validation of Corruption Patterns

Visual inspection was performed to verify that synthetic masks behaved as intended. Validation confirmed that stripe corruption disrupted multiple retinal layers simultaneously and preserved surrounding anatomical context. Inspection across disease categories showed that stripe masking remained effective under healthy anatomy and irregular pathological deformation.

6 Classical Inpainting Baselines

Motivation for Baseline Comparisons

Before evaluating diffusion-based reconstruction, it is necessary to establish classical baseline methods. Baselines provide interpretable reference points for determining whether a

generative model actually improves reconstruction quality. In retinal OCT imaging, local anatomical continuity is often strong enough that simple deterministic methods may already perform well. Consequently, a diffusion model should not only produce visually plausible reconstructions, but also outperform simpler interpolation-based approaches. The baseline methods considered in this project range from trivial reconstruction to more sophisticated PDE-inspired inpainting. These methods provide insight into the role of local smoothness assumptions and the limitations of purely deterministic reconstruction.

6.1 Zero-Fill Reconstruction

The simplest baseline is zero-fill reconstruction. In this approach, missing pixels are not reconstructed at all. The corrupted image itself is treated as the final output:

$$\hat{X}_{\text{zero}} = Y = M \odot X.$$

Although this method performs no inference, it provides a lower bound for reconstruction quality. Any meaningful reconstruction method should improve substantially over zero-fill masking.

6.2 Gaussian Blur Reconstruction

Gaussian reconstruction estimates missing pixels using local smoothing. The corrupted image is convolved with a Gaussian kernel:

$$G_{\sigma}(x, y) = \frac{1}{2\pi\sigma^2} \exp\left(-\frac{x^2 + y^2}{2\sigma^2}\right).$$

The reconstructed image is then $\hat{X}_{\text{gauss}} = G_{\sigma} * Y$, where $*$ denotes convolution.

Gaussian smoothing propagates neighboring intensity values into the missing region and partially restores continuity across the stripe mask. However, the method assumes local smoothness and therefore oversmooths sharp retinal boundaries.

6.3 PDE-Based Diffusion Inpainting

PDE-based inpainting reconstructs missing regions using diffusion equations that propagate neighboring intensity structure inward. A common isotropic diffusion formulation is:

$$\frac{\partial u}{\partial t} = \Delta u,$$

where Δ denotes the Laplacian operator. The method iteratively smooths intensity variation while preserving continuity between neighboring pixels. Inpainting proceeds by evolving the image over artificial time until the missing region is filled.

PDE-based reconstruction generally performs better than simple Gaussian smoothing because information propagates adaptively from surrounding structure rather than through uniform convolution alone. However, the method remains fundamentally local. Large missing regions require long-range anatomical inference, which PDE diffusion cannot model effectively. In retinal OCT images, PDE methods may preserve coarse continuity while still failing to reconstruct realistic retinal layer geometry.

6.4 Telea Fast Marching Inpainting

Telea inpainting reconstructs missing pixels using a fast marching method that propagates information inward from mask boundaries. Instead of relying purely on isotropic smoothing, Telea reconstruction prioritizes nearby boundary structure and local gradient information. Pixels closer to known regions are reconstructed first, and their values influence subsequent reconstruction steps.

However, Telea reconstruction still relies primarily on local structure propagation and may fail when large anatomical regions are missing or when pathology introduces substantial ambiguity.

Qualitative Comparison of Classical Methods

Visual inspection revealed clear differences between the baseline methods. Zero-fill reconstruction leaves a sharp black stripe across the retinal image and completely disrupts retinal continuity. Gaussian smoothing partially restores continuity but produces blurry reconstructions with oversmoothed boundaries. Thin retinal structures often become indistinct. PDE-based diffusion preserves continuity more effectively than Gaussian blur, particularly for gradual anatomical transitions. However, reconstructed regions may still appear overly smooth and lack detailed retinal structure. Telea inpainting produced the most visually coherent classical reconstructions. Retinal layers often extended more naturally across masked regions, and boundary continuity was preserved more effectively.

Despite these improvements, all classical methods struggled when corruption removed larger portions of retinal anatomy. None of the deterministic baselines could reliably reconstruct fine pathological detail or globally consistent retinal geometry in difficult cases.

Quantitative Baseline Performance

Baseline performance was evaluated using MSE, MAE, and PSNR. Among the classical methods, Telea inpainting achieved the strongest overall performance. PDE-based diffusion generally outperformed Gaussian smoothing, while zero-fill reconstruction produced the weakest results.

Representative evaluation results were:

Telea: $\text{MSE} \approx 0.011$, $\text{PSNR} \approx 28.6$,

PDE: $\text{MSE} \approx 0.032$,

Gaussian: $\text{MSE} \approx 0.135$.

These results demonstrate that relatively simple deterministic methods can already perform surprisingly well on structured OCT inpainting tasks. This observation becomes especially important later when evaluating the initial diffusion model, which initially underperformed several classical baselines despite substantially greater model complexity.

7 Diffusion-Based Inpainting

Motivation for Generative Reconstruction

Classical inpainting methods reconstruct missing regions primarily through local interpolation and deterministic smoothing. While these approaches can preserve continuity in simple settings, they struggle when large anatomical regions are removed or when retinal structure becomes highly nonlinear. Retinal OCT reconstruction is fundamentally ambiguous. Multiple plausible anatomical completions may exist for the same corrupted image. Consequently, reconstruction can be interpreted probabilistically rather than deterministically.

Generative models attempt to learn the underlying image distribution

$$p_{\text{data}}(x)$$

directly from retinal scans. Instead of filling pixels using local heuristics alone, a generative model learns statistical structure from the dataset itself. Diffusion models are especially attractive because they provide stable generative training, support iterative reconstruction, and naturally model uncertainty through stochastic sampling.

The goal of DiffIndEye is therefore to study whether a conditional diffusion model can reconstruct anatomically plausible retinal OCT structure from masked observations.

Overview of Denoising Diffusion Models

Denoising diffusion probabilistic models generate images through iterative denoising. The framework consists of a forward process that gradually corrupts images with Gaussian noise, and a reverse process that learns to remove this noise.

Starting from a clean image x_0 , the forward diffusion process constructs a sequence $x_1, x_2, \dots, x_T$ that progressively approaches Gaussian noise. The reverse model then learns $p_{\theta}(x_{t-1} | x_t)$,

allowing images to be reconstructed step-by-step from noisy states. Unlike GANs, diffusion models optimize a stable denoising objective and explicitly model probabilistic transitions between diffusion states.

7.1 Forward Noise Process

The forward diffusion process gradually corrupts an image using Gaussian noise:

$$q(x_t | x_{t-1}) = \mathcal{N}\left(\sqrt{1 - \beta_t}x_{t-1}, \beta_t I\right),$$

where β_t is the variance schedule, and I is the identity covariance matrix. Repeated application of this process produces increasingly noisy images. After sufficiently many steps,

$$x_T \approx \mathcal{N}(0, I).$$

A useful closed-form representation is:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon,$$

where:

$$\varepsilon \sim \mathcal{N}(0, I),$$

and

$$\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s).$$

This formulation allows noisy training samples to be generated efficiently without simulating every intermediate diffusion step explicitly.

7.2 Reverse Denoising Process

The reverse process attempts to reconstruct the clean image by removing Gaussian noise iteratively. The model learns:

$$p_{\theta}(x_{t-1} | x_t),$$

which approximates the reverse diffusion dynamics. Rather than predicting the clean image directly, most diffusion implementations predict the added noise:

$$\varepsilon_{\theta}(x_t, t).$$

The training objective becomes:

$$\mathcal{L} = \mathbb{E} \left[\|\varepsilon - \varepsilon_{\theta}(x_t, t)\|^2 \right].$$

During sampling, the model repeatedly denoises:

$$x_T \rightarrow x_{T-1} \rightarrow \cdots \rightarrow x_0.$$

This iterative refinement process allows diffusion models to generate highly structured images while preserving training stability.

7.3 Conditional Diffusion Framework

DiffIndEye uses conditional diffusion rather than unconditional image generation. Instead of generating arbitrary retinal OCT scans, the model reconstructs missing regions conditioned on the observed image. Conditioning constrains the reconstruction process so that observed retinal structure remains fixed while missing regions are reconstructed probabilistically. This transforms diffusion from a pure image synthesis problem into a structured inverse problem.

Conditioning on Masked OCT Images

The diffusion model receives three primary inputs: X_t, Y, M . These tensors are concatenated along the channel dimension before convolutional processing: $Z = [X_t, Y, M]$. The conditioning image provides anatomical context surrounding the missing stripe region, while the mask explicitly identifies which pixels are corrupted.

Time Conditioning and Positional Embeddings

Because the reverse process depends on diffusion timestep t , the model must receive information about the current noise level. Diffusion timesteps are encoded using sinusoidal positional embeddings:

$$\gamma(t) = (\sin(\omega_1 t), \cos(\omega_1 t), \dots, \sin(\omega_k t), \cos(\omega_k t)).$$

These embeddings provide a continuous representation of diffusion time and allow the model to adapt its denoising behavior depending on the current noise scale. The timestep embedding is projected through a small multilayer perceptron before being injected into convolutional feature representations. Early diffusion states require small denoising corrections, while later states require large-scale structural reconstruction from nearly pure noise.

Initial Encoder–Decoder Architecture

The initial DiffIndEye model uses a lightweight convolutional encoder–decoder architecture consisting of convolutional encoding layers, intermediate latent feature processing, and convolutional decoding layers.

The baseline architecture processes (128 x 256) retinal tensors and contains approximately 115,000 trainable parameters. ReLU nonlinearities are used throughout the network:

$$\sigma(x) = \max(0, x).$$

The architecture was intentionally kept simple in order to validate the reconstruction pipeline and test diffusion scheduling.

Current Architectural Limitations

Although the initial diffusion framework trains successfully, reconstruction quality remains limited. The baseline architecture underperforms several classical inpainting methods despite successfully optimizing the diffusion objective.

One major limitation is the absence of a U-Net structure. The current encoder–decoder architecture lacks skip connections, multi-scale feature propagation, and hierarchical spatial fusion. As a result, high-frequency retinal detail is frequently lost during reconstruction, particularly along thin retinal interfaces and sharp reflectivity boundaries.

The model also operates entirely in pixel space without latent compression, attention mechanisms, or multi-resolution feature representations. Consequently, the network struggles to preserve globally coherent retinal geometry while simultaneously reconstructing fine anatomical structure.

These architectural constraints likely contribute to several observed reconstruction artifacts, including excessive smoothing, blurry retinal layers, weak structural continuity, and incomplete boundary preservation across masked regions.

The results therefore motivate future extensions involving U-Net diffusion architectures, attention-based conditioning, improved sampling procedures, and more expressive multi-scale feature hierarchies capable of preserving both local retinal detail and long-range anatomical consistency.

7.4 (WIP) U-Net Diffusion

Preliminary experiments were extended using a lightweight U-Net-based conditional diffusion architecture. The primary motivation for introducing the U-Net structure was to improve preservation of high-frequency retinal anatomy through skip connections and multi-scale feature propagation.

The resulting architecture contains approximately 475,000 trainable parameters, substantially larger than the initial baseline DDPM. Early experiments demonstrated significantly improved optimization behavior during training. The U-Net diffusion model achieved substantially lower diffusion loss values relative to the original encoder–decoder architecture.

However, quantitative reconstruction metrics improved only marginally:

$$\text{MSE}_{\text{baseline}} \approx 0.340, \quad \text{MSE}_{\text{U-Net}} \approx 0.329.$$

These results suggest that improved denoising optimization does not necessarily translate directly into improved anatomical reconstruction, and that retinal OCT inpainting likely requires stronger spatial inductive bias beyond shallow skip connections alone. The current U-Net implementation therefore serves primarily as an intermediate architectural milestone establishing that the modular DiffIndEye framework can support scalable diffusion experimentation beyond the original baseline DDPM.

8 Training Framework

Training Objectives

The primary objective of the diffusion model is to learn how to reconstruct retinal OCT structure from noisy and partially observed inputs. Rather than predicting the clean image directly, the model is trained to estimate the Gaussian noise added during the forward diffusion process.

Given a clean retinal image x_0 , a diffusion timestep t and sampled Gaussian noise

$$\varepsilon \sim \mathcal{N}(0, I),$$

the noisy image is generated as

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon.$$

The model then learns a mapping

$$\varepsilon_\theta(x_t, t, Y, M)$$

that predicts the noise component. The overall goal is therefore to approximate the reverse denoising process well enough that retinal structure can be reconstructed iteratively from noisy masked observations.

Noise Prediction Formulation

The baseline DiffIndEye implementation uses the standard DDPM noise prediction objective:

$$\mathcal{L} = \mathbb{E} \left[\|\varepsilon - \varepsilon_\theta(x_t, t, Y, M)\|^2 \right].$$

Instead of predicting the clean image directly, the model predicts the noise that was added during diffusion. This formulation improves numerical stability and simplifies optimization.

The noisy image x_t is sampled directly using the closed-form diffusion equation:

$$x_t = \sqrt{\bar{\alpha}_t}x_0 + \sqrt{1 - \bar{\alpha}_t}\varepsilon.$$

At each training iteration:

1. a retinal image is sampled,
2. a corruption mask is applied,
3. a timestep t is sampled uniformly,
4. Gaussian noise is generated,
5. and the model predicts the noise realization.

Masked Loss Strategy

The baseline implementation optionally supports a masked reconstruction loss. Instead of computing error across the entire image, the loss may focus primarily on corrupted regions:

$$\mathcal{L}_{\text{masked}} = \|(1 - M) \odot (\varepsilon - \varepsilon_\theta)\|^2.$$

This encourages the model to prioritize reconstruction quality inside the missing stripe region rather than allocating excessive capacity toward already visible anatomy. Without masked weighting, the optimization objective may become dominated by observed retinal structure because most image pixels remain uncorrupted. The masked loss therefore increases emphasis on the actual inpainting task. In practice, both masked and full-image objectives can be useful. Full-image denoising encourages global consistency, while masked denoising emphasizes missing-region reconstruction.

Diffusion Scheduling

The diffusion schedule controls how noise is added throughout the forward process. DiffIndEye uses a linear variance schedule:

$$\beta_t \in [\beta_{\min}, \beta_{\max}],$$

with:

$$\beta_{\min} = 10^{-4}, \quad \beta_{\max} = 2 \times 10^{-2}.$$

The cumulative signal coefficient is:

$$\bar{\alpha}_t = \prod_{s=1}^t (1 - \beta_s).$$

The baseline implementation uses 1000 diffusion steps. Early timesteps preserve substantial retinal structure, while later timesteps approach nearly pure Gaussian noise. The schedule plays a critical role in training stability. If noise grows too rapidly, the model receives almost no structural information at later timesteps. If noise grows too slowly, the reverse process may not learn sufficiently robust denoising behavior.

Optimization Procedure

The model is trained using stochastic gradient descent with minibatches of retinal OCT images. Optimization uses the Adam optimizer:

$$\theta \leftarrow \theta - \eta \nabla_{\theta} \mathcal{L},$$

where θ denotes network parameters, and η is the learning rate.

Typical training configurations included:

$$\text{batch size} = 8, \quad \text{learning rate} = 10^{-4}.$$

Training proceeds iteratively across sampled retinal images and diffusion timesteps. At each iteration:

1. a batch is loaded,
2. masks are generated,
3. noisy diffusion states are sampled,
4. the model predicts noise,
5. and gradients are updated.

Training was performed initially on CPU hardware, which significantly increased iteration time and motivated later refactoring for more efficient experimentation.

Checkpointing and Experiment Tracking

Checkpointing was introduced to support reproducible experimentation and iterative model development. During training, model states are periodically saved. Each checkpoint contains:

- model parameters,
- optimizer state,
- configuration settings,

- and training metadata.

Checkpoints are stored using timestamped run directories:

`run_name_YYYYMMDD_HHMMSS.`

This structure allows comparison between diffusion runs, rollback to earlier experiments, and hyperparameter tuning. The training logs also track loss trajectories and checkpoint timing. These procedures became increasingly important as the project evolved from exploratory notebooks into a reusable research framework.

9 Evaluation Methodology

Evaluation Objectives

The evaluation framework was designed to compare classical inpainting methods against diffusion-based reconstruction under controlled synthetic corruption. The primary objective was to determine whether the diffusion model could reconstruct retinal OCT structure more effectively than deterministic baselines.

Evaluation focused on both:

- quantitative reconstruction accuracy,
- and qualitative anatomical plausibility.

Because retinal OCT reconstruction is structurally constrained, numerical accuracy alone is insufficient. Reconstructions must also preserve:

- retinal layer continuity,
- reflectivity transitions,
- and global anatomical organization.

The evaluation pipeline therefore combined pixel-wise metrics with visual inspection across multiple reconstruction methods and checkpoints.

9.1 Mean Squared Error

Mean squared error measures the average squared difference between reconstructed and ground-truth intensities:

$$\text{MSE} = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W (X_{ij} - \hat{X}_{ij})^2.$$

Lower MSE values indicate closer numerical agreement between the reconstruction and the original retinal image. Because squaring penalizes large deviations heavily, MSE is sensitive to strong reconstruction artifacts, boundary discontinuities, and large intensity errors. Although MSE is widely used, it does not necessarily correlate perfectly with anatomical realism. Two reconstructions may have similar MSE values while exhibiting very different retinal structure visually.

9.2 Mean Absolute Error

Mean absolute error measures the average absolute reconstruction difference:

$$\text{MAE} = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W |X_{ij} - \hat{X}_{ij}|.$$

Unlike MSE, MAE penalizes errors linearly rather than quadratically. As a result, it is less sensitive to isolated large reconstruction deviations. MAE provides a complementary measure of reconstruction fidelity and is often easier to interpret directly because it reflects average pixel-level reconstruction error.

9.3 Peak Signal-to-Noise Ratio

Peak signal-to-noise ratio measures reconstruction quality relative to the maximum possible image intensity:

$$\text{PSNR} = 10 \log_{10} \left(\frac{L^2}{\text{MSE}} \right),$$

where L is the maximum pixel intensity.

Because retinal OCT images were normalized into (0,1) the metric simplifies to:

$$\text{PSNR} = 10 \log_{10} \left(\frac{1}{\text{MSE}} \right).$$

Higher PSNR values indicate better reconstruction quality. The metric is commonly used in image restoration because it summarizes reconstruction fidelity on a logarithmic scale. However, PSNR still remains a pixel-wise metric and may not fully capture anatomical realism or structural continuity.

Masked-Region Evaluation

Although reconstruction metrics can be computed across the full image, the primary region of interest is the corrupted stripe mask.

Let:

$$M \in \{0, 1\}^{H \times W}$$

denote the binary mask. The masked reconstruction error is computed only over pixels satisfying: $M_{ij} = 0$. For example, masked MSE becomes:

$$\text{MSE}_{\text{masked}} = \frac{\sum_{i,j} (1 - M_{ij}) (X_{ij} - \hat{X}_{ij})^2}{\sum_{i,j} (1 - M_{ij})}.$$

Masked evaluation is important because full-image metrics may be dominated by already observed retinal structure. Focusing on the missing region isolates the actual inpainting problem. This evaluation strategy provides a more meaningful comparison between reconstruction methods.

Baseline-vs-Diffusion Comparisons

The first evaluation workflow compares the diffusion model against classical baselines including:

- zero-fill reconstruction,
- Gaussian smoothing,
- PDE-based diffusion,
- and Telea inpainting.

Each method reconstructs identical masked retinal images under the same corruption settings. Quantitative metrics are then averaged across evaluation samples:

$$\{X_i\}_{i=1}^N.$$

This comparison establishes whether the diffusion model improves upon deterministic interpolation approaches. The initial experiments revealed that classical methods, especially Telea inpainting, performed surprisingly well relative to the baseline diffusion model. This result became one of the major motivations for later architectural improvements.

Diffusion-vs-Diffusion Comparisons

The second evaluation workflow compares different diffusion checkpoints against one another.

Given checkpoints:

$$\theta_1, \theta_2, \dots, \theta_k,$$

the framework evaluates reconstruction quality across different training runs, hyperparameter settings, and architecture variants. Diffusion-vs-diffusion evaluation became especially important after the project was refactored into a modular experimental framework.

Qualitative Visualization Pipeline

Quantitative metrics alone are insufficient for evaluating retinal OCT reconstruction because low numerical error does not necessarily imply anatomically realistic structure. Consequently, visual inspection was incorporated throughout the evaluation pipeline.

For each reconstruction method, comparisons included the original retinal image, the masked image, and the reconstructed output. These visualizations allowed assessment of retinal layer continuity, boundary preservation, anatomical plausibility, and reconstruction artifacts. The visualization pipeline revealed several important failure modes in the initial diffusion model, including blurry retinal structure, weak high-frequency reconstruction, and incomplete anatomical continuity. In contrast, Telea inpainting often preserved retinal layers surprisingly well despite its relatively simple deterministic formulation.

These observations reinforced the importance of combining numerical evaluation with visual and anatomical interpretation when studying retinal OCT reconstruction.

10 Experimental Results

Classical Baseline Performance

The classical reconstruction methods produced substantially different reconstruction quality under stripe corruption. Zero-fill reconstruction performed worst because the masked region remained entirely missing. Gaussian smoothing improved continuity but introduced substantial blurring and loss of retinal detail. PDE-based diffusion reconstruction preserved anatomical continuity more effectively and reduced visible stripe artifacts. Among all classical methods, Telea inpainting achieved the strongest overall performance, preserving retinal layer continuity remarkably well despite its relatively simple deterministic formulation.

Method	Mean MSE	Mean MAE	Mean PSNR
Telea Inpainting	0.011	0.071	28.6
PDE Diffusion	0.032	0.151	22.8
Gaussian Blur	0.135	0.352	16.1
Zero-Fill	0.169	0.398	15.1

Table 1: Quantitative reconstruction performance of classical inpainting baselines.

Initial Diffusion Training Results

The initial diffusion model trained successfully from an optimization perspective. The DDPM objective decreased steadily during training, and the model learned stable denoising behavior

across diffusion timesteps. Training was performed using $T = 1000$ diffusion steps with $\beta_{\min} = 10^{-4}$ and $\beta_{\max} = 2 \times 10^{-2}$. The baseline architecture contained approximately 115,169 trainable parameters and operated on 128×256 retinal OCT tensors. Checkpointing and training diagnostics confirmed that tensor dimensions remained stable, diffusion schedules behaved correctly, and loss values remained finite throughout optimization. Despite stable training behavior, reconstruction quality remained relatively weak compared to the classical baselines, with representative diffusion results yielding $\text{MSE} \approx 0.340$ and $\text{PSNR} \approx 12.0$. These results suggest that although the model learned the diffusion objective successfully, the baseline architecture was insufficient for strong anatomical retinal reconstruction.

Training Stability Analysis

Training stability improved substantially after refactoring the project into a modular framework. Early notebook-based experimentation occasionally produced tensor mismatch issues, checkpoint inconsistencies, and unstable evaluation workflows. After refactoring, the pipeline supported deterministic preprocessing, standardized checkpointing, reproducible evaluation, and automated experiment tracking. Sanity checks confirmed that $\bar{\alpha}_0 \approx 1$ and $\bar{\alpha}_T \approx 0$, indicating that the diffusion schedule behaved correctly throughout training. Although optimization remained stable and losses decreased smoothly without catastrophic divergence, stable denoising behavior did not necessarily correspond to strong anatomical reconstruction. The model frequently learned to reduce diffusion loss numerically while still producing anatomically weak retinal reconstructions.

Reconstruction Quality Analysis

Visual reconstruction quality varied substantially across methods. Telea inpainting consistently preserved retinal layer continuity better than the diffusion model, while PDE diffusion also produced relatively coherent anatomical structure in many cases. In contrast, the baseline DDPM frequently generated blurry retinal layers, weak boundary reconstruction, and incomplete anatomical continuity across the masked stripe region. Although the diffusion outputs often removed the visible corruption artifact itself, reconstructed regions lacked detailed retinal structure and frequently appeared oversmoothed relative to surrounding anatomy. These results suggest that the baseline architecture learned coarse denoising behavior successfully but failed to capture sufficiently rich retinal representations for high-quality anatomical reconstruction.

Failure Cases and Artifacts

Several recurring failure modes were observed during diffusion reconstruction. The most common issue was excessive smoothing, which caused loss of thin retinal boundaries, sharp reflectivity transitions, and fine anatomical detail. The model also struggled with long-range

retinal continuity, with some reconstructions exhibiting misaligned retinal layers across the corrupted region. Additional artifacts included blurry stripe interiors, inconsistent intensity scaling, and noisy reconstruction textures. These limitations likely arise from the shallow encoder–decoder architecture, lack of skip connections, insufficient multi-scale feature propagation, and limited spatial context modeling. The experiments therefore suggest that optimizing the diffusion objective alone is insufficient for strong OCT reconstruction without architectures specifically designed to preserve detailed anatomical structure.

Comparison Against Classical Methods

One of the most important findings of the project was that both diffusion implementations underperformed several classical baselines despite substantially greater model complexity. Telea inpainting achieved the strongest overall reconstruction quality, preserving retinal layer continuity remarkably well under stripe corruption. In contrast, both the initial DDPM and the U-Net diffusion model produced substantially higher reconstruction error and weaker anatomical consistency. Although the U-Net architecture improved optimization behavior and reduced diffusion training loss considerably, the quantitative reconstruction metrics improved only marginally relative to the original encoder–decoder DDPM. These results suggest that successful diffusion optimization alone is insufficient for strong retinal OCT reconstruction and that architectural design, spatial feature propagation, and reconstruction inductive bias remain critical components of anatomically meaningful inpainting.

Method	Mean MSE	Mean MAE	Mean PSNR
Telea Inpainting	0.011	0.071	28.58
PDE Diffusion	0.032	0.151	22.81
Gaussian Blur	0.135	0.352	16.14
Zero-Fill	0.169	0.398	15.15
Initial DDPM	0.340	0.572	12.01
U-Net DDPM	0.329	0.563	12.14

Table 2: Quantitative comparison of classical inpainting methods, the initial DDPM architecture, and the U-Net diffusion model on retinal OCT stripe reconstruction.

11 Discussion

Interpretation of Experimental Findings

The experiments demonstrate that retinal OCT inpainting is substantially more difficult than standard image interpolation tasks. Although the diffusion framework trained successfully from an optimization perspective, the resulting reconstructions frequently lacked fine anatomical detail and coherent retinal structure.

At the same time, several classical methods performed unexpectedly well. In particular, Telea inpainting preserved retinal continuity far more effectively than anticipated given its comparatively simple deterministic formulation.

These findings suggest that retinal OCT images possess strong local geometric regularity. Because retinal layers are spatially smooth and horizontally continuous, local propagation methods can often reconstruct moderate corruption regions effectively without requiring highly expressive generative models.

The experiments therefore highlight an important distinction between:

successful denoising optimization $\neq$ successful anatomical reconstruction.

A model may minimize diffusion loss while still failing to preserve clinically meaningful retinal structure.

Why Classical Methods Currently Outperform Diffusion

The initial diffusion model underperformed several classical baselines despite substantially greater model complexity. This result appears surprising initially, but several factors help explain the outcome.

First, the baseline DDPM architecture is relatively shallow and lacks mechanisms for preserving high-resolution spatial detail. Classical methods such as Telea inpainting directly propagate local edge and boundary information, which is especially effective for horizontally structured retinal anatomy. Second, the reconstruction task itself remains relatively local under the current stripe corruption setup. Since surrounding retinal structure remains visible, deterministic propagation methods can often interpolate the missing region effectively without requiring sophisticated generative reasoning. Third, diffusion models are highly dependent on architecture quality. The diffusion objective alone does not guarantee strong image reconstruction performance. If the network lacks sufficient representational capacity or spatial inductive bias, the learned denoising process may become overly smooth and anatomically imprecise.

The current results therefore do not imply that diffusion reconstruction is fundamentally ineffective for OCT imaging. Rather, they indicate that the present architecture is insufficiently expressive relative to the complexity of the reconstruction problem.

12 Future Directions

12.1 Improved Sampling Procedures

The current implementation uses standard DDPM sampling with iterative reverse denoising across $T = 1000$ diffusion steps. Although stable, this procedure is computationally expensive

and may accumulate reconstruction error over long sampling trajectories. Future work will explore accelerated methods such as DDIM sampling, higher-order diffusion solvers, and consistency-based reconstruction approaches to improve both sampling efficiency and reconstruction quality.

12.2 Alternative Noise Schedules

The current framework uses a linear variance schedule for the forward diffusion process. Future work may investigate cosine, sigmoid, or learned variance schedules to improve training stability and reconstruction fidelity. Because the noise schedule controls the signal-to-noise ratio throughout diffusion, alternative schedules may better preserve retinal structure at intermediate diffusion states and improve denoising behavior during reconstruction.

12.3 Attention Mechanisms and Multi-Scale Features

The baseline architecture relies primarily on local convolutional processing. However, retinal OCT reconstruction often requires long-range anatomical reasoning because stripe corruption interrupts multiple retinal layers simultaneously.

Attention mechanisms may help the model capture global spatial relationships:

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}}\right)V.$$

Self-attention layers allow the model to relate distant retinal regions directly rather than relying only on local convolutional propagation.

Future architectures may combine:

- convolutional feature extraction,
- attention-based global context modeling,
- and multi-scale diffusion representations.

Such architectures may improve reconstruction of:

- long retinal boundaries,
- pathological deformation,
- and global retinal geometry.

12.4 Expanded Corruption Strategies

The current framework uses structured horizontal stripe masking as a controlled reconstruction benchmark for retinal OCT inpainting. Although effective for studying retinal continuity, this corruption pattern remains relatively simple compared to real imaging artifacts. Future work may explore more realistic corruption schemes including multiple stripe masks, irregular geometric corruption, localized artifact simulation, and pathology-aware masking. These extensions could better approximate motion artifacts, vascular shadowing, signal attenuation, and partial acquisition failure commonly observed in retinal imaging, while also helping evaluate whether diffusion models can generalize beyond highly structured synthetic corruption patterns.

12.5 Uncertainty Quantification

One advantage of diffusion models is their probabilistic nature. Because reconstruction proceeds through stochastic sampling, multiple plausible retinal reconstructions may be generated for the same masked image.

Future work may therefore study uncertainty explicitly by sampling:

$$\hat{X}^{(1)}, \hat{X}^{(2)}, \dots, \hat{X}^{(K)} \sim p(X | Y, M).$$

The resulting sample variance could be used to estimate reconstruction uncertainty:

$$\text{Var}(\hat{X}_{ij}).$$

Uncertainty estimation is especially important in medical imaging because ambiguous regions may correspond to clinically important pathology. Rather than producing a single deterministic reconstruction, probabilistic models may provide confidence estimates regarding reconstructed anatomy.

Uncertainty maps may also help identify:

- unreliable reconstructions,
- anatomically ambiguous regions,
- and regions requiring clinician review.

12.6 Application to Real Retinal Artifacts

The current framework evaluates reconstruction under synthetic stripe corruption because synthetic masks provide controlled benchmarking and known ground truth. Future work

may extend diffusion reconstruction to real retinal OCT artifacts including motion distortion, vascular shadowing, signal attenuation, and acquisition dropout. Because real artifacts often overlap with pathology and lack known ground-truth anatomy, future pipelines may combine artifact detection, segmentation, and conditional reconstruction within a unified restoration framework.

12.7 Potential Clinical and Research Applications

Although the current project is exploratory, diffusion-based retinal reconstruction may eventually support applications including OCT artifact correction, image quality enhancement, probabilistic anatomical reconstruction, uncertainty-aware imaging, and data augmentation for retinal analysis pipelines. More broadly, DiffIndEye contributes to the growing intersection between medical image analysis, probabilistic generative modeling, and computational ophthalmology. The current experiments also highlight that successful medical image reconstruction depends not only on diffusion objectives themselves, but also on architectures capable of preserving anatomically meaningful spatial structure.

13 Conclusion

Summary of Findings

This project studied retinal OCT inpainting using both classical image processing methods and diffusion-based generative reconstruction. Classical approaches, particularly Telea inpainting, established strong reconstruction baselines and consistently outperformed the initial diffusion implementations. Although the DDPM models trained stably and successfully optimized the denoising objective, reconstruction quality remained limited, especially for preserving fine retinal anatomy and long-range structural continuity. These results suggest that retinal OCT inpainting depends heavily on architectural design and spatial feature representation in addition to the diffusion framework itself.

Contributions of DiffIndEye

DiffIndEye contributes a reproducible experimental framework for studying diffusion-based retinal OCT reconstruction. The project developed a standardized preprocessing pipeline, synthetic corruption framework, classical reconstruction baselines, conditional DDPM implementations, and modular training and evaluation workflows. The resulting infrastructure supports rapid experimentation and provides a foundation for future research involving U-Net diffusion architectures, improved sampling procedures, attention mechanisms, and uncertainty-aware reconstruction.

Broader Implications for Medical Image Inpainting

The project highlights several broader themes in medical image analysis and generative modeling. First, anatomically meaningful reconstruction requires more than low numerical reconstruction error alone. Second, relatively simple deterministic methods may remain highly competitive for structured medical imaging tasks. Finally, diffusion models remain promising for medical imaging because they provide probabilistic and uncertainty-aware reconstruction capabilities, but their success depends critically on architectures capable of preserving detailed anatomical structure. DiffIndEye therefore establishes an initial foundation for future work at the intersection of medical image analysis, computer vision, and probabilistic generative modeling.